Imbalanced Text Classification Study

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Group Number: 11

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1 Introduction

This project focuses on binary classification of disaster-related tweets using machine learning models. Due to class imbalance, special emphasis is given to improving minority class (Class-1) performance using various techniques.

2 Dataset Description

The dataset contains approximately 11,000 tweets labeled as:

- Class 0: Non-disaster tweets
- Class 1: Disaster tweets

The dataset is imbalanced, with fewer disaster tweets compared to non-disaster tweets.

3 Preprocessing

- Missing text values were handled.
- Text was converted to numerical features using TF-IDF.
- Data was split into training, validation, and test sets.

4 Methodology

The following models were implemented:

- Logistic Regression
- Random Forest
- Neural Network (MLP)

Techniques applied:

- Hyperparameter tuning (GridSearchCV)
- Threshold tuning
- Imbalance handling:
  - Class weighting
  - SMOTE

5 Experimental Setup

- Train-test split: 80-20
- Validation split: 10% from training data
- Cross-validation: 5-fold (Logistic Regression)
- Metrics:
  - Accuracy
  - Precision, Recall, F1-score
  - ROC-AUC
  - PR-AUC

6 Results

6.1 Logistic Regression

Best Parameters: C = 10, Penalty = L2, Solver = lbfgs

Before Threshold Tuning

- Accuracy: 0.88
- Class 1: Precision = 0.74, Recall = 0.54, F1 = 0.62

Best Threshold: 0.394

After Threshold Tuning

- Accuracy: 0.87
- Class 1: Precision = 0.68, Recall = 0.62, F1 = 0.65

After Class Weighting

- Accuracy: 0.83
- Class 1: Precision = 0.53, Recall = 0.76, F1 = 0.62

After SMOTE

- Accuracy: 0.84
- Class 1: Precision = 0.56, Recall = 0.74, F1 = 0.64

ROC-AUC: 0.89
PR-AUC: 0.71

6.2 Random Forest

Best Parameters: n_estimators = 200, max_depth = None, min_samples_split = 5, min_samples_leaf = 1

Before Threshold Tuning

- Accuracy: 0.87
- Class 1: Precision = 0.76, Recall = 0.45, F1 = 0.57

Best Threshold: 0.361

After Threshold Tuning

- Accuracy: 0.86
- Class 1: Precision = 0.63, Recall = 0.63, F1 = 0.63

After Class Weighting

- Accuracy: 0.85
- Class 1: Precision = 0.58, Recall = 0.71, F1 = 0.64

After SMOTE

- Accuracy: 0.85
- Class 1: Precision = 0.60, Recall = 0.68, F1 = 0.64

ROC-AUC: 0.888
PR-AUC: 0.710

6.3 Neural Network (MLP)

Best Configuration: Activation = Tanh, Optimizer = Adam, LR = 0.001, Init = Glorot Uniform

Before Imbalance Handling

- Accuracy: 0.86
- Class 1: Precision = 0.62, Recall = 0.58, F1 = 0.60

Best Threshold: 0.521

After Class Weighting

- Accuracy: 0.82
- Class 1 Recall: 0.77

After SMOTE

- Accuracy: 0.85
- Class 1: Precision = 0.61, Recall = 0.59, F1 = 0.60

ROC-AUC: 0.855
PR-AUC: 0.650

7 Analysis and Discussion

- Logistic Regression performs best overall due to its effectiveness on high-dimensional TF-IDF features.
- Random Forest improves significantly after threshold tuning and imbalance handling.
- MLP underperforms compared to other models due to difficulty in handling sparse features.
- Class weighting increases recall but reduces precision.
- SMOTE provides a better balance between precision and recall.

8 Conclusion

Logistic Regression is the most suitable model for this task. Random Forest provides competitive performance, while MLP is less effective. Among imbalance handling techniques, SMOTE achieves the best trade-off, while class weighting maximizes recall.